

Nmap Cheat Sheet

Host Discovery

Command	Description	Related Switches
nmap -sn 192.168.1.0/24	Ping scan (disable port scan)	-sn: No port scan
nmap -Pn 192.168.1.1	Skip ping (assume host is up)	-Pn: No ping
nmap -PS 192.168.1.1	TCP SYN ping	-PS[port-list]: Can specify custom ports
nmap -PA 192.168.1.1	TCP ACK ping	-PA[port-list]: Can specify custom ports
nmap -PU 192.168.1.1	UDP ping	-PU[port-list]: Can specify custom ports
nmap -PE 192.168.1.1	ICMP echo ping	-PE: ICMP echo request (ping)
nmap -PP 192.168.1.1	ICMP timestamp ping	-PP: ICMP timestamp request
nmap -PM 192.168.1.1	ICMP address mask ping	-PM: ICMP address mask request

Port Scanning

Command	Description	Related Switches
nmap 192.168.1.1	Default scan (top 1000 TCP ports)	Default behavior without specific flags
nmap -p 80,443 192.168.1.1	Scan specific ports	-p: Specify ports to scan
nmap -p 1-100 192.168.1.1	Scan port range	-p: Supports range notation
nmap -p- 192.168.1.1	Scan all 65535 ports	-p-: All ports
nmap -F 192.168.1.1	Fast scan (top 100 ports)	-F: Fast mode
nmap -top-ports 2000 192.168.1.1	Scan top 2000 common ports	-top-ports <n>: Scan n most common ports
nmap -p http,https 192.168.1.1	Scan named ports	-p: Supports service names from /etc/services
nmap -T0 192.168.1.1	Paranoid timing (slowest)	-T<0-5>: Timing template (0=slowest, 5=fastest)
nmap -T5 192.168.1.1	Insane timing (fastest)	-T<0-5>: Timing template (0=slowest, 5=fastest)

Scan Types

Command	Description	Related Switches
nmap -sS 192.168.1.1	TCP SYN scan (default, need root)	-sS: SYN scan (stealthy)
nmap -sT 192.168.1.1	TCP connect scan (no root required)	-sT: Full TCP connect scan
nmap -sU 192.168.1.1	UDP scan	-sU: UDP scan
nmap -sA 192.168.1.1	TCP ACK scan	-sA: ACK scan for firewall rule mapping
nmap -sW 192.168.1.1	TCP Window scan	-sW: TCP Window scan
nmap -sM 192.168.1.1	TCP Maimon scan	-sM: Maimon scan
nmap -sN 192.168.1.1	TCP Null scan	-sN: No flags set
nmap -sF 192.168.1.1	TCP FIN scan	-sF: FIN flag only
nmap -sX 192.168.1.1	TCP Xmas scan	-sX: FIN, PSH and URG flags set
nmap -sl zombie:port 192.168.1.1	Idle/Zombie scan	-sl <zombie:port>: Idle scan through zombie host
nmap -sO 192.168.1.1	IP protocol scan	-sO: Protocol scan
nmap -b ftp-bounce:21 192.168.1.1	FTP bounce scan	-b <FTP relay host>: FTP bounce scan
nmap -sY 192.168.1.1	SCTP INIT scan	-sY: SCTP INIT scan
nmap -sZ 192.168.1.1	SCTP COOKIE-ECHO scan	-sZ: SCTP COOKIE-ECHO scan

Version and OS Detection

Command	Description	Related Switches
nmap -sV 192.168.1.1	Version detection	-sV: Probe open ports for service/version info
nmap -sV --version-intensity 0-9	Set version intensity	--version-intensity <0-9>: From light to aggressive probing
nmap -sV --version-light	Light version detection (intensity 2)	--version-light: Limit to most likely probes (intensity 2)
nmap -sV --version-all	Try all probes (intensity 9)	--version-all: Try every probe (intensity 9)
nmap -A 192.168.1.1	Enable OS detection, version, script scanning, traceroute	-A: Aggressive scan options
nmap -O 192.168.1.1	OS detection	-O: Enable OS detection
nmap -O --osscan-limit	Limit OS detection to promising targets	--osscan-limit: Limit OS detection to promising targets
nmap -O --osscan-guess	More aggressive OS detection	--osscan-guess: Guess OS more aggressively
nmap -O --max-os-tries 1	Limit OS detection tries	--max-os-tries <n>: Set maximum number of OS detection tries



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Output Formats

Command	Description
<code>nmap -oN output.txt 192.168.1.1</code>	Normal output
<code>nmap -oX output.xml 192.168.1.1</code>	XML output
<code>nmap -oG output.grep 192.168.1.1</code>	Grepable output
<code>nmap -oA output 192.168.1.1</code>	Output in all formats
<code>nmap -oS output.scr 192.168.1.1</code>	Script kiddie output
<code>nmap -v 192.168.1.1</code>	Verbose output
<code>nmap -vv 192.168.1.1</code>	Very verbose output
<code>nmap --reason 192.168.1.1</code>	Display reason a port is in a particular state
<code>nmap --open 192.168.1.1</code>	Only show open ports
<code>nmap --packet-trace 192.168.1.1</code>	Show all packets sent and received
<code>nmap -d 192.168.1.1</code>	Debugging
<code>nmap --iflist</code>	List available interfaces and routes

NSE (Nmap Scripting Engine)

Command	Description
<code>nmap -sC 192.168.1.1</code>	Default scripts
<code>nmap --script default 192.168.1.1</code>	Default scripts
<code>nmap --script=banner 192.168.1.1</code>	Banner script
<code>nmap --script=http-* 192.168.1.1</code>	All HTTP scripts
<code>nmap --script=vuln 192.168.1.1</code>	Vulnerability assessment
<code>nmap --script=safe 192.168.1.1</code>	Safe scripts
<code>nmap --script=auth 192.168.1.1</code>	Authentication scripts
<code>nmap --script=discovery 192.168.1.1</code>	Discovery scripts
<code>nmap --script=malware 192.168.1.1</code>	Malware detection scripts
<code>nmap --script=version 192.168.1.1</code>	Version detection scripts
<code>nmap --script=updatedb</code>	Update script database
<code>nmap --script-help=script.nse</code>	Get help for a script

Timing and Performance

Command	Description	Related Switches
<code>nmap -T0 192.168.1.1</code>	Paranoid: Very slow, evades IDS	-T0: Serial, 5 min between scans
<code>nmap -T1 192.168.1.1</code>	Sneaky: Slow, evades IDS	-T1: Serial, 15 sec between scans
<code>nmap -T2 192.168.1.1</code>	Polite: Slows down to consume less bandwidth	-T2: Serial, 0.4 sec between scans
<code>nmap -T3 192.168.1.1</code>	Normal: Default speed	-T3: Parallel scanning
<code>nmap -T4 192.168.1.1</code>	Aggressive: Fast, assumes reliable network	-T4: More aggressive parallel scanning
<code>nmap -T5 192.168.1.1</code>	Insane: Very fast, assumes extremely reliable network	-T5: Insanely aggressive scanning
<code>nmap --min-rate 100 192.168.1.1</code>	Send packets no slower than 100 per second	--min-rate <rate>: Send packets no slower than <rate> per second
<code>nmap --max-rate 100 192.168.1.1</code>	Send packets no faster than 100 per second	--max-rate <rate>: Send packets no faster than <rate> per second
<code>nmap --max-retries 2 192.168.1.1</code>	Limit retry attempts for port scans	--max-retries <tries>: Cap number of port scan probe retransmissions
<code>nmap --min-parallelism 10</code>	Min probe parallelization	--min-parallelism <n>: Min probe parallelization
<code>nmap --max-parallelism 10</code>	Max probe parallelization	--max-parallelism <n>: Max probe parallelization
<code>nmap --min-hostgroup 50</code>	Min hosts scanned in parallel	--min-hostgroup <n>: Min hosts per scan group
<code>nmap --max-hostgroup 100</code>	Max hosts scanned in parallel	--max-hostgroup <n>: Max hosts per scan group
<code>nmap --scan-delay 1s</code>	Adjust delay between probes	--scan-delay <time>: Adjust delay between probes
<code>nmap --max-scan-delay 10s</code>	Max scan delay	--max-scan-delay <time>: Max scan delay

Firewall / IDS Evasion and Spoofing

Command	Description	Related Switches
<code>nmap -f 192.168.1.1</code>	Fragment packets	-f: Fragment packets (may bypass filters)
<code>nmap --mtu 24 192.168.1.1</code>	Specify MTU size	--mtu <n>: Use specified MTU for fragmentation
<code>nmap -D decoy1,decoy2 192.168.1.1</code>	Cloak scan with decoys	-D <decoy1,decoy2,...>: Mask scan using decoys
<code>nmap -S source_ip 192.168.1.1</code>	Spoof source IP address	-S <IP_Address>: Spoof source address
<code>nmap -e eth0 192.168.1.1</code>	Use specified interface	-e <interface>: Use specified interface
<code>nmap --source-port 53 192.168.1.1</code>	Use given port number as source port	--source-port <port>: Use custom source port
<code>nmap --data-length 200 192.168.1.1</code>	Append random data to packets	--data-length <n>: Append random data to packets
<code>nmap --randomize-hosts 192.168.1.1</code>	Randomize target host order	--randomize-hosts: Randomize target host order
<code>nmap --spooof-mac MAC 192.168.1.1</code>	Spoof MAC address	--spooof-mac <MAC>: Spoof MAC address
<code>nmap --proxies proxy_url 192.168.1.1</code>	Use HTTP/SOCKS4 proxies	--proxies <url1,[url2],...>: Relay connections through HTTP/SOCKS4 proxies
<code>nmap --badsum 192.168.1.1</code>	Send packets with bogus checksums	--badsum: Send packets with bogus TCP/UDP checksums
<code>nmap --ttl 60 192.168.1.1</code>	Set IP time-to-live field	--ttl <val>: Set IP time-to-live field

Advanced Target Specification

Command	Description	Related Switches
<code>nmap 192.168.1.1 192.168.2.1</code>	Scan multiple targets	Multiple IP addresses space separated
<code>nmap 192.168.1.0/24</code>	Scan entire subnet	CIDR notation for network blocks
<code>nmap 192.168.1.1-254</code>	Scan a range	Hyphen notation for ranges
<code>nmap scanme.nmap.org</code>	Scan domain	Hostnames are resolved via DNS
<code>nmap -iL targets.txt</code>	Scan targets from file	-iL <file>: Input from list of hosts/networks
<code>nmap -iR 10</code>	Scan 10 random hosts	-iR <num>: Choose random targets
<code>nmap --exclude 192.168.1.1</code>	Exclude hosts	--exclude <host1,[host2],...>: Exclude hosts/networks
<code>nmap --excludefile exclude.txt</code>	Exclude hosts from file	--excludefile <file>: Exclude list from file

